AidFlow - Initial System Design

Introduction

AidFlow is a platform for managing donations, allowing organizations to streamline the donation process and centralize community support. The system enables the linking of donors, campaign coordinators, volunteers and storage personnel on a single platform.

The primary goal of AidFlow is to streamline the donation process by arranging donation offers, keeping track of available items, and coordinating collection and delivery efforts.

The platform is designed to minimize confusion, avoid double allocation of donated goods and promote efficient distribution of donated goods.

System Overview

AidFlow provides a centralized platform where different users can participate in donation campaigns.

The system supports five main user roles:

1. **Donors:** Offer items to donation campaigns.
2. **Campaign Coordinators:** Create campaigns, review donation offers, and monitor progress.
3. **Volunteers:** Collect and deliver donated items.
4. **Storage Staff:** Record received donations and manage inventory.
5. **Administrators:** Manage user accounts and oversee the platform.

The system keeps donation information organized so that users can track items from the initial offer until final delivery.

Initial System Architecture

AidFlow will use a proposed three-layer architecture to separate the user interface, application logic, and data storage.

Presentation Layer

The presentation layer represents the interface that users interact with.

It will include:

- Login and registration pages
- User dashboard
- Donation campaign pages
- Donation offer forms
- Inventory management pages
- Collection and delivery tracking pages

The interface will be designed to be simple and easy to navigate.

Application Layer

The application layer manages the main operations of AidFlow.

Its responsibilities include:

- User authentication and role management
- Campaign creation and management
- Donation offer processing
- Donation status tracking
- Inventory management
- Volunteer task coordination
- Donation allocation and delivery tracking

This layer processes user requests and applies to the system's rules.

Data Layer

The data layer stores and organizes information required by the system.

It will contain:

- User account information
- Donation campaign records

- Donation offers
- Inventory records
- Volunteer assignments
- Collection and delivery records

The data layer allows the application to retrieve and update information when necessary.

Initial System Workflow

The proposed workflow of AidFlow is:

1. A campaign coordinator creates a donation campaign.
2. A donor views the campaign and submits a donation offer.
3. The coordinator reviews and accepts or rejects the offer.
4. A volunteer collects the accepted donation.
5. Storage staff confirm receipt and update the inventory.
6. The coordinator allocates available items to the campaign.
7. Volunteers deliver the allocated donations.
8. The system records the delivery status.

This workflow helps ensure that donations are tracked throughout the process.

Initial User Interface Design

The initial interface will focus on simplicity and accessibility.

Main Screens

1. Login and Registration Screen

Allow users to create accounts and securely log in.

2. Dashboard

Displays relevant information based on the user's role, such as active campaigns, donation status, and assigned tasks.

3. Campaign Management Screen

Allow coordinators to create campaigns and manage requested items.

4. Donation Offer Screen

Allows donors to submit information about items they wish to donate.

5. Inventory Screen

Allow storage staff to record received items and view available quantities.

6. Delivery Tracking Screen

Allow volunteers and coordinators to monitor collection and delivery progress.

Conclusion

The first design for AidFlow is a good start to a donation management platform. The proposed architecture is a separation of the user interface, application operations, and data storage to maintain a logical organization of the system. The first design can be refined once the functional and non-functional requirements are determined.